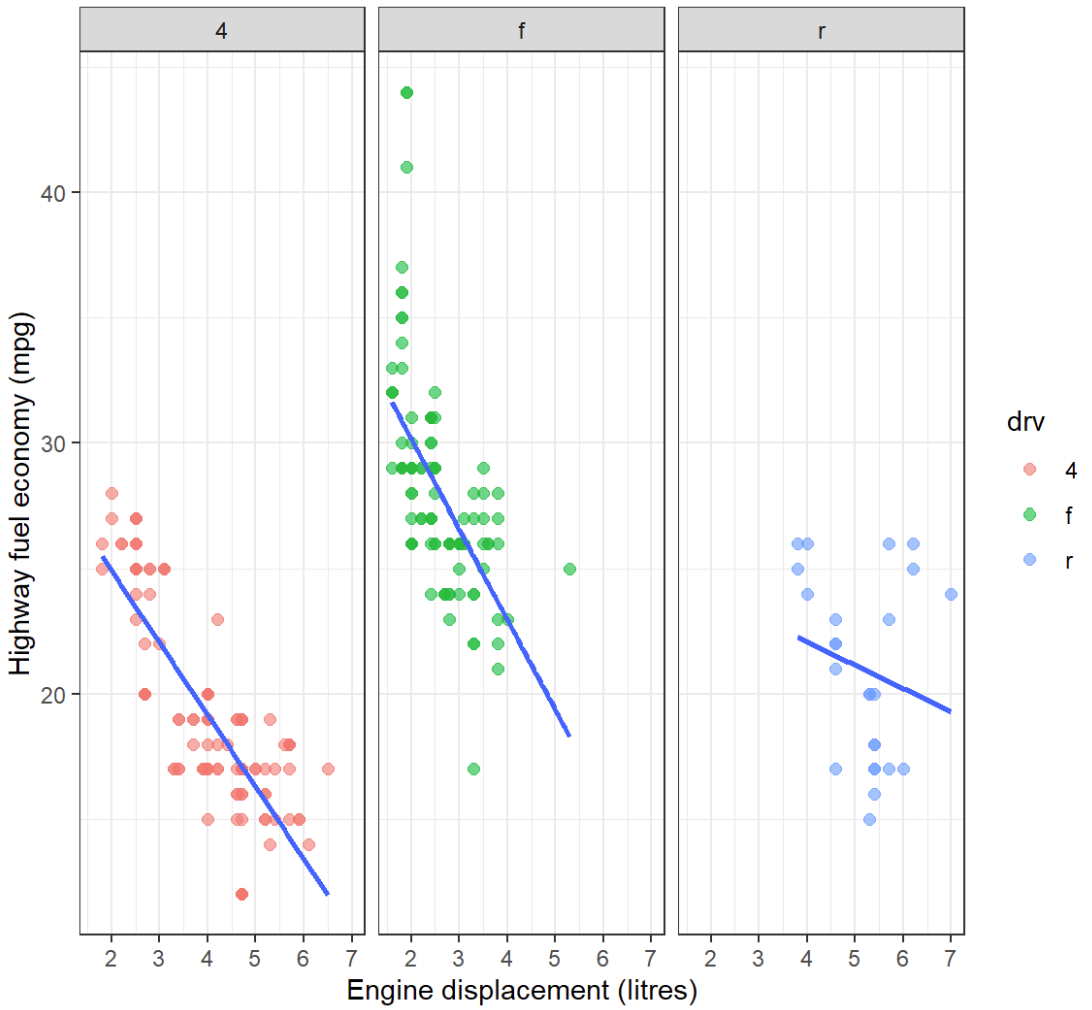
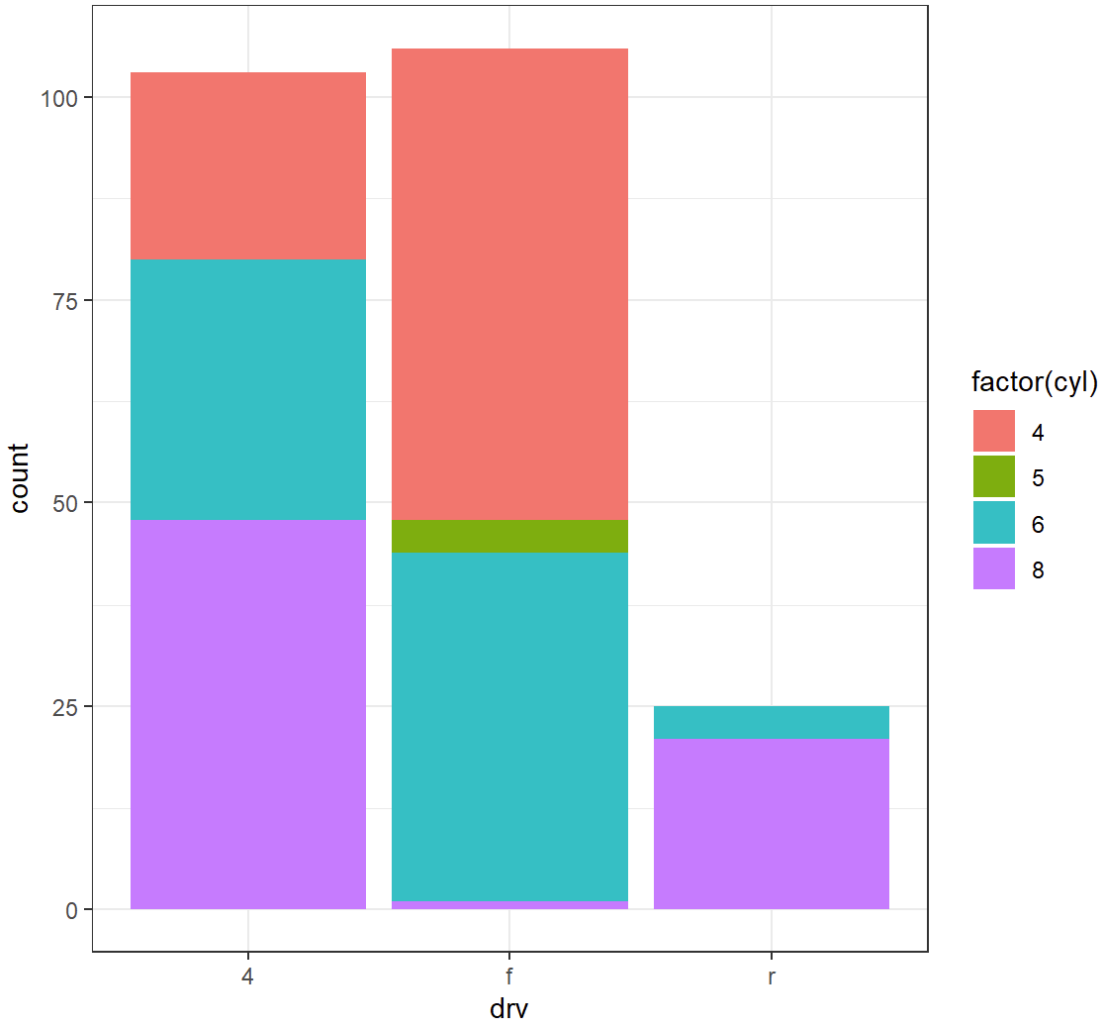
Homework 3

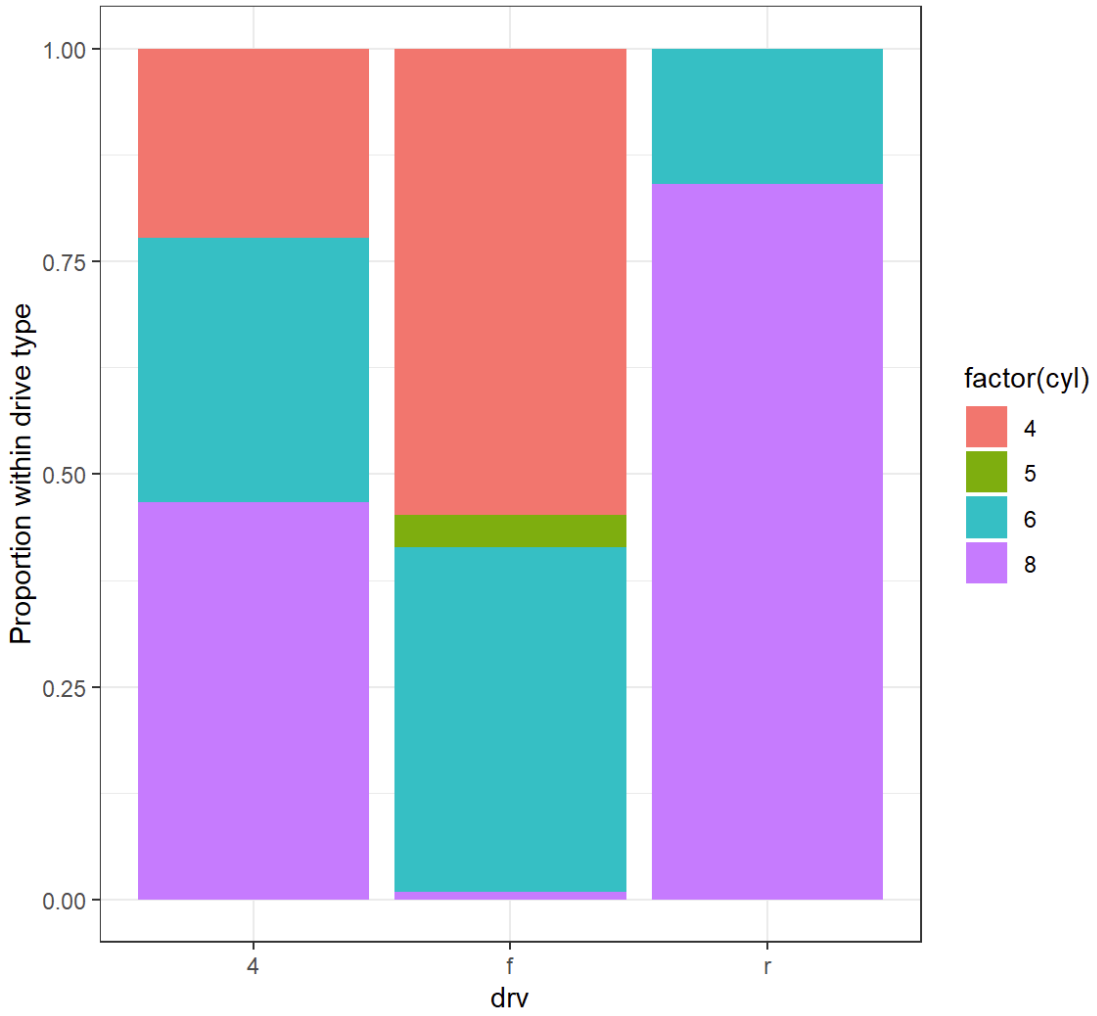
Problem 1

**Part A.** There are three lines because facet_wrap splits the data into three panels (4, f, r), and geom_smooth is fit separately in each. Color only affects the points.

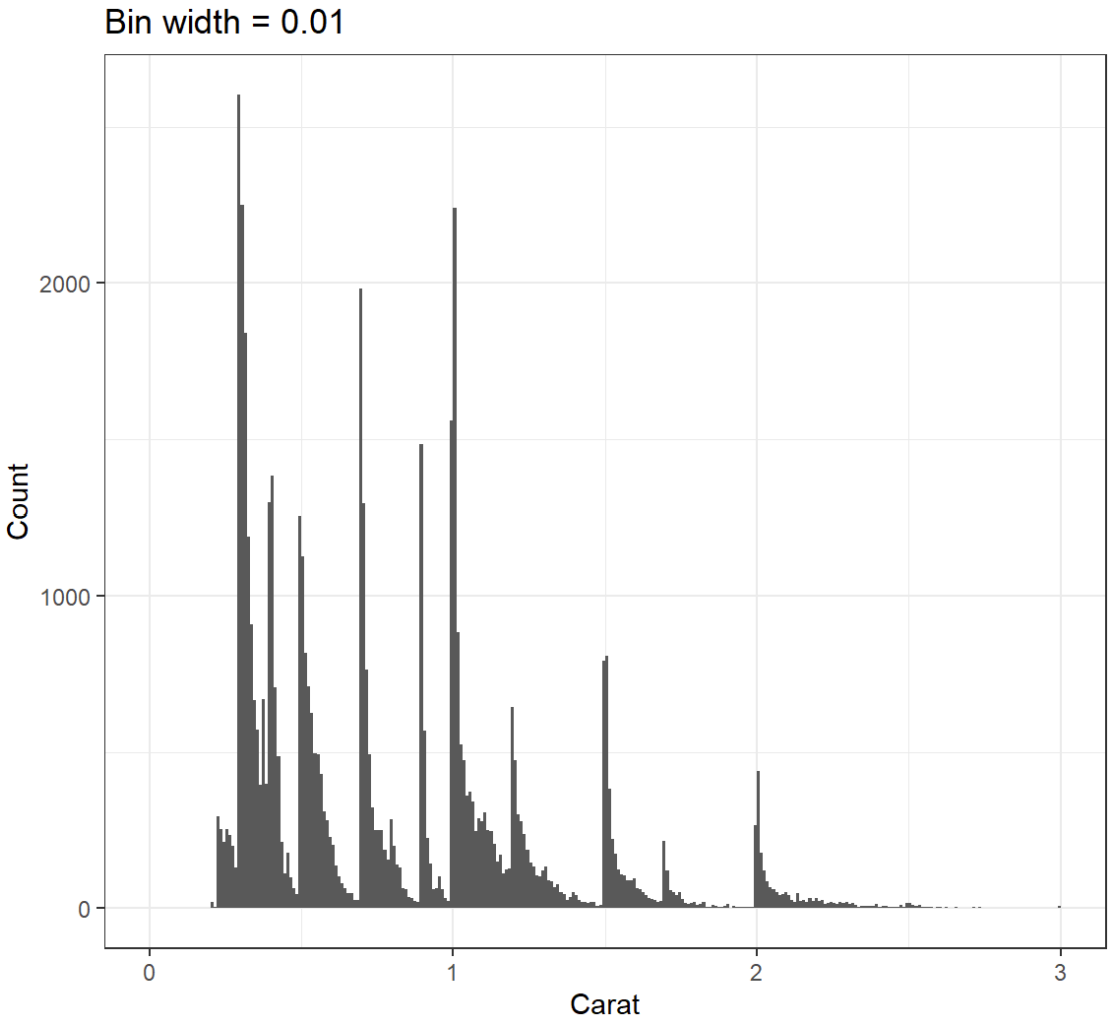


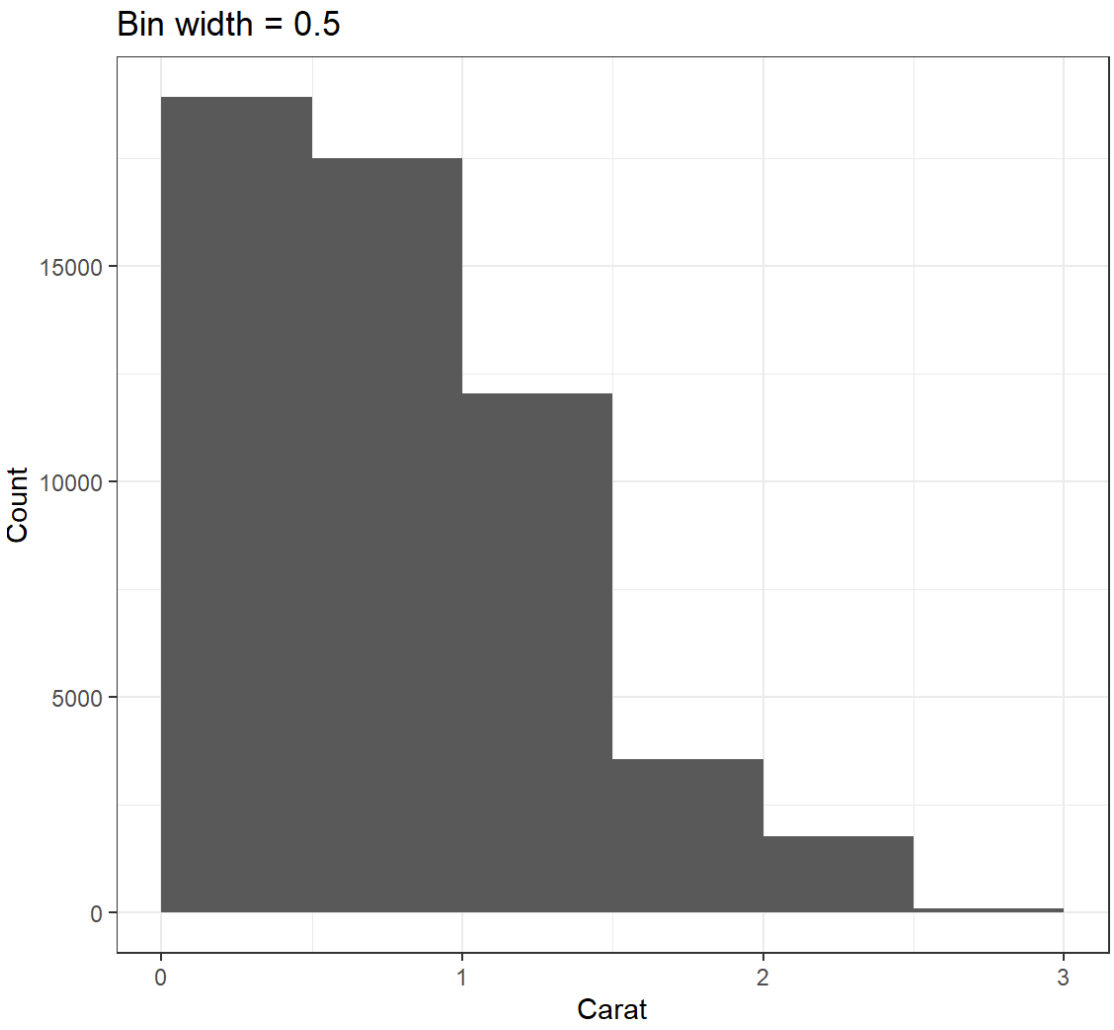

**Part B.** In the right plot every bar is scaled to 1, so it shows each drive type's cylinder mix and hides how many cars there are. In the count plot, 4 and f have about 100 cars each and r has about 25.


100
75
50
25
0
count
4
f
r
drv
factor(cyl)
4
5
6
8



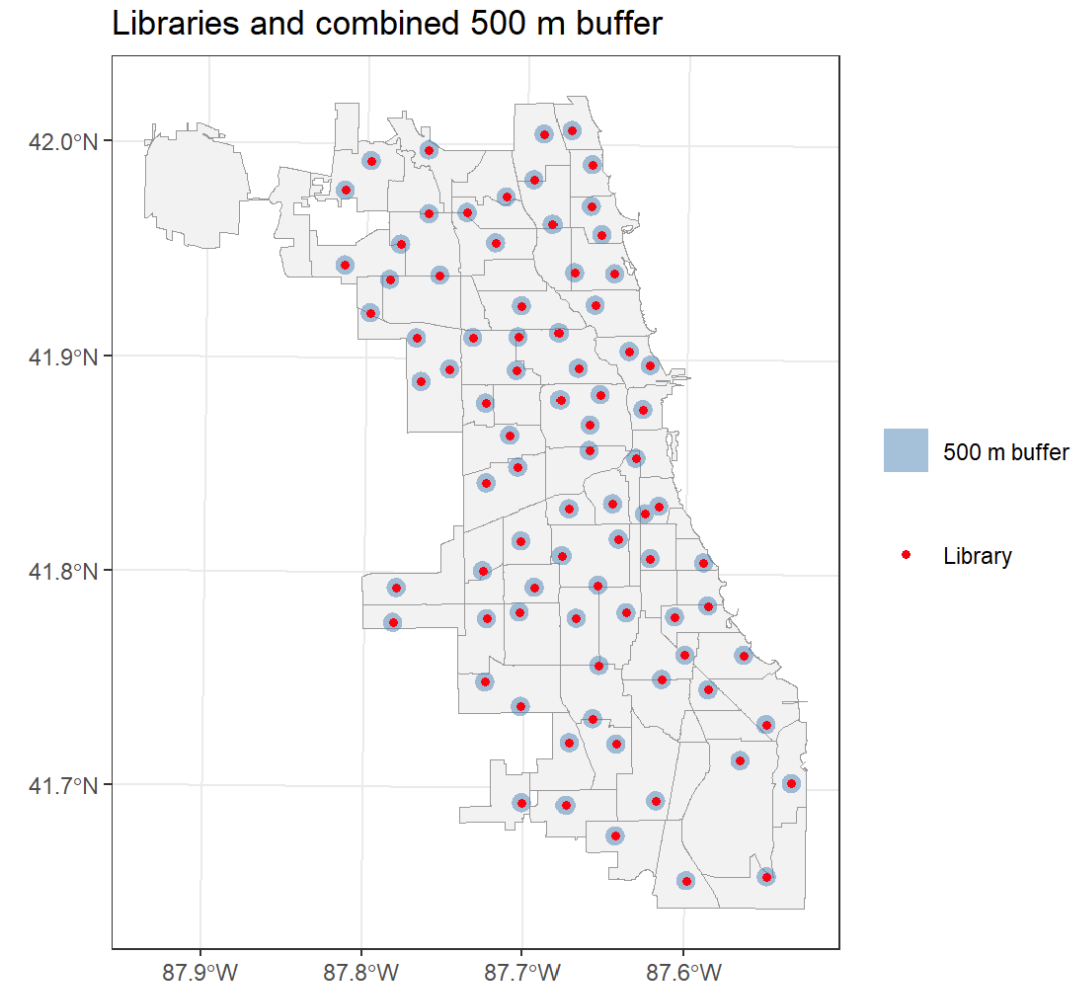

**Part C.** Kept 53,908 records (carat $\leqslant$ 3) and excluded 32. There are 23 diamonds at exactly 0.99 carat and 1,558 at exactly 1.00. The narrow bins show spikes at round sizes like 0.5 and 1.0, and almost no 0.99 stones next to a huge pile at 1.00, so sizes seem to be rounded up. The wide bins hide this.


Bin width = 0.01
Count
2000
1000
0
0
1
2
3
Carat




Problem 2

**Data.** Libraries and community area boundaries both come from the City of Chicago Data Portal, accessed Nov 8, 2026, and are recorded in data_sources.csv. The raw files were not changed.



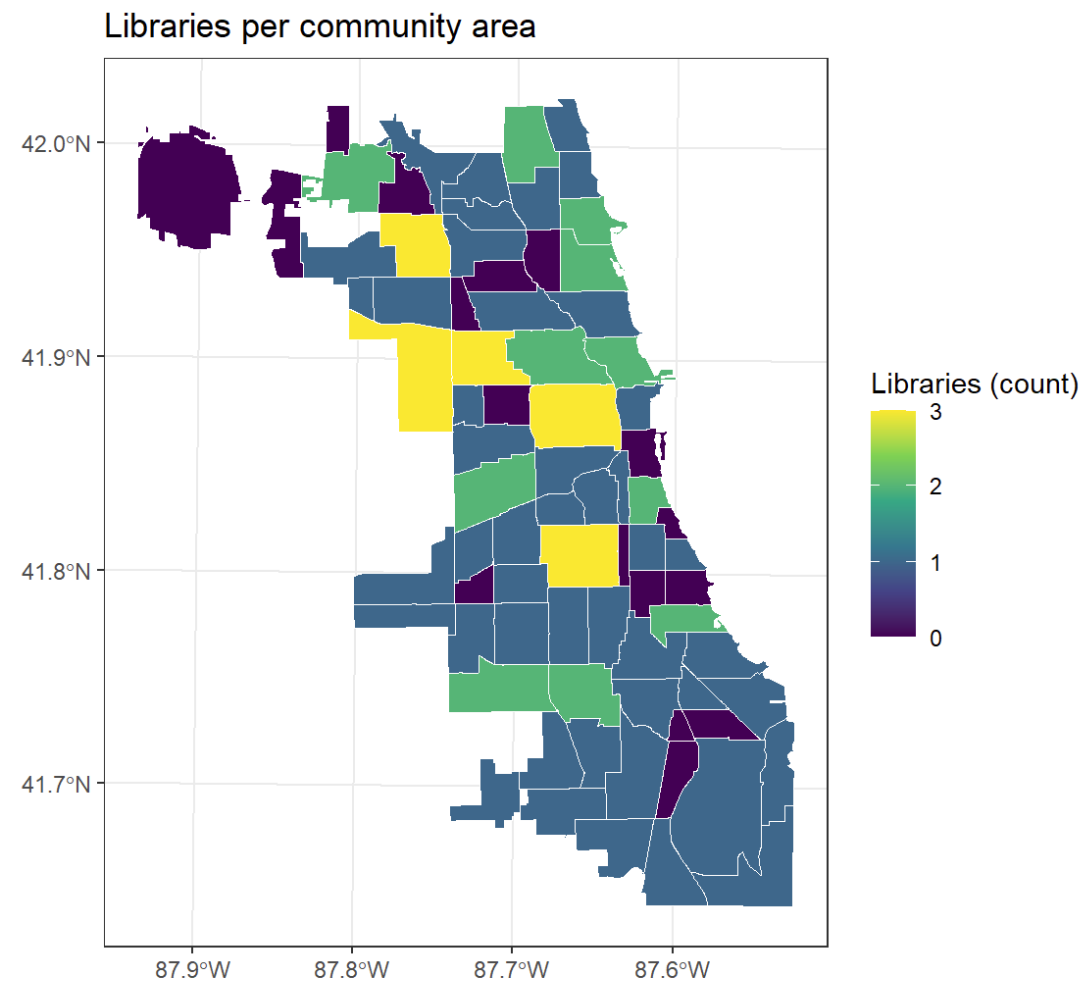

urce: City of Chicago Data Portal (libraries, community areas), accessed Nov 8 2026

**Cleaning.** 82 records came in and 82 were kept. There were 0 bad coordinates, 0 duplicate names and 0 shared coordinates, so nothing was removed. Points are saved as sites_m in NAD83 / UTM zone 16N (EPSG:26916), in meters.

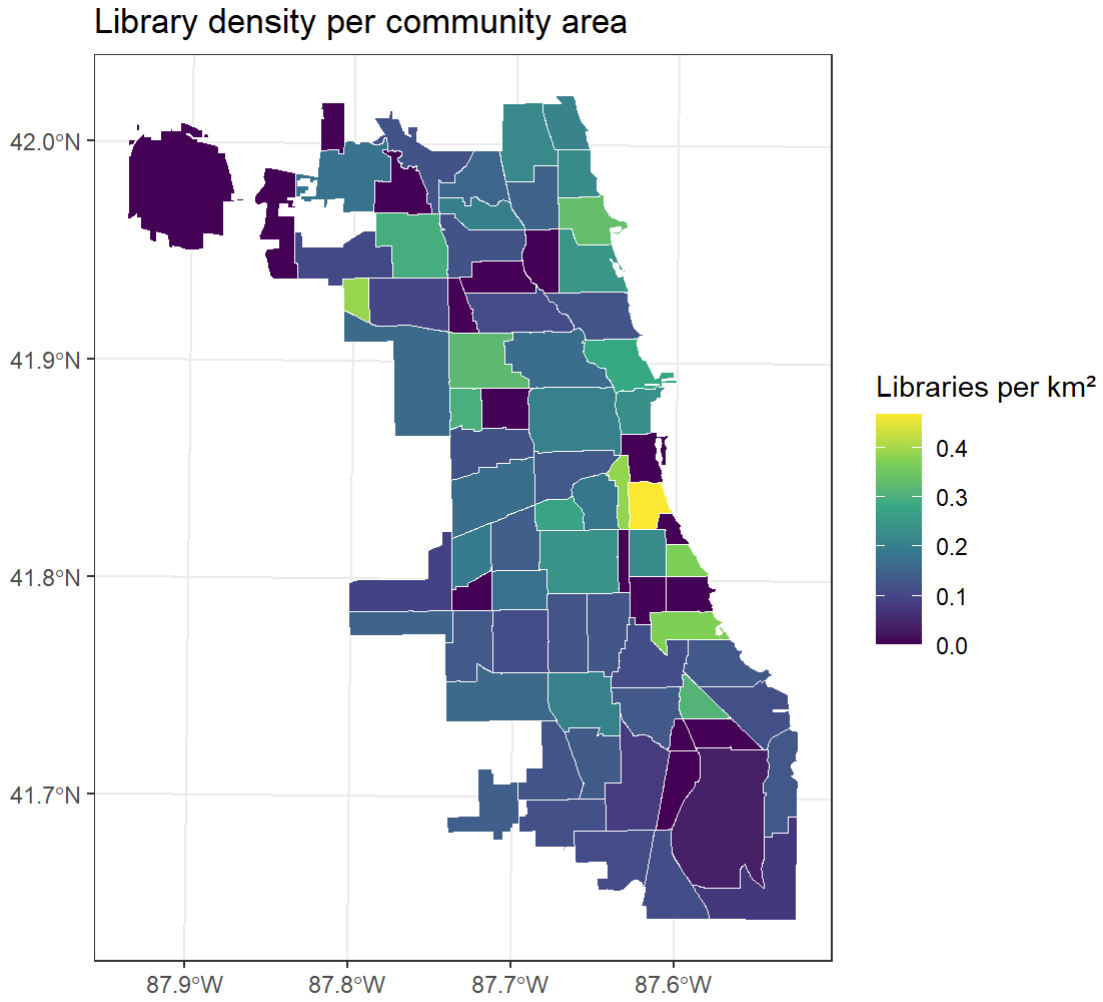
| radius_m | union_area_km2 |
|---|---|
| 500 | 64.3 |
| 1000 | about 250 |

**Interpretation.** About 64 km² is within 500 m of a library and about 250 km² is within 1,000 m. That is roughly 3.9 times more, just under 4 times because overlapping circles are counted once. Some buffer area falls outside the city boundary and I kept it.

**Problem 3**




Source: City of Chicago Data Portal, accessed Nov 8 2026. Sites left unassigned: 0




Source: City of Chicago Data Portal, accessed Nov 8 2026. Sites left unassigned: 0

**Boundary checks.** There are 77 community areas with unique IDs and names. I found 0 duplicate IDs, 0 empty shapes and 0 invalid shapes, so no repairs were needed. They are in EPSG:26916, the same CRS as sites_m.

**Match audit.** All 82 sites matched exactly one area. 0 matched zero areas, 0 matched more than one, and 0 sites were left unassigned.

**Results.** 16 of 77 areas have zero libraries, and I kept them. Counts add up to 82. The most libraries by count is 3, in New City, Humboldt Park, Austin, Near West Side and Portage Park. The highest densities are Douglas (0.468 per km²), Montclare (0.391), Armour Square (0.388), Woodlawn (0.373) and Kenwood (0.371).

**Interpretation.** The comparison changes when area size is included. Austin is tied for the most libraries (3), but it is large (18.5 km²), so its density is only 0.162 per km². Douglas has only 2 libraries but is small (4.27 km²), which gives the highest density. Raw counts favor large areas

and density favors small ones.